

GTU CSE 241 / 501 Quiz

Std Code:1440

```
#ifndef example-h
#define example-h
#include <iostream>
using namespace std;

class Time {
public:
    int hours;
    int minutes;

    Time (int h, int m) {hours = h; minutes = m;}
    // for initialize
    hours = 0;
    minutes = 0;

    void printTime() {
        cout << hours << ":" << minutes << endl;
    }

    void setHours (int new-h) {
        if (new-h >= 0 && new-h < 24) {
            hours = new-h;
        } else {
            cerr << "Invalid value!" << endl;
        }
    }

    void setMinutes (int new-m) {
        if (new-m >= 0 && new-m < 60) {
            minutes = new-m;
        } else {
            cerr << "Invalid value!" << endl;
        }
    }

    void addMinutes (int add) {
        if (add + minutes >= 60) {
            int y = add + minutes;
            int x = y % 60;
            minutes = x;
            if (hours + x >= 24) { hours = (hours + x) % 24; }
        }
    }

    bool isEqual (Time t2) {
        if (hours == t2.hours && minutes == t2.minutes) {
            return true;
        } else {
            return false;
        }
    }

    bool isEarlier (Time t2) {
        if (hours > t2.hours) {
            return true;
        } if (hours == t2.hours) {
            if (minutes > t2.minutes) {
                return true;
            } else {
                return false;
            }
        }
    }

    int getMinutes () {
        return minutes;
    }
};
```

```
#include "library.h"
```

```
int main() {
```

```
    Time t1(14,45);
```

```
    Time t2(9);
```

```
    Time t3;
```

```
    t1.printTime();
```

```
    if (t1.isEqual(t2) || t1.isEarlier(t2)) {
```

```
        cout << "t1 == t2 or t1 < t2 \n";
```

```
    t1.addMinutes(80);
```

```
    t2.setHours(33);
```

```
    t2.setMinutes(t1.getMinutes());
```

```
    return 0;
```

```
}
```